

CoVR-2: Automatic Data Construction for Composed Video Retrieval

January 13, 2025

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Abstract

*Composed Image Retrieval (CoIR) has gained significant attention for enabling the retrieval of target images from vast datasets using a combination of a reference image and a textual modification. Composed Video Retrieval (CoVR) [1] extends this paradigm by introducing the ability to retrieve videos based on both visual and textual cues, significantly enhancing search capabilities for dynamic and complex content. CoVR-2 introduces a scalable methodology for automatic dataset construction tailored for CoVR, addressing the limitations of manual annotations in CoIR. By leveraging the BLIP-2 architecture and integrating a novel consistency loss function, CoVR-2 achieves state-of-the-art results on CIRR [2] and CIRCO datasets. **Our Code***



Figure 1: Illustration of Composed Video Retrieval

1. Introduction

In an era of information abundance, retrieving relevant content efficiently has become a challenge across various domains. Traditional search methods, which often rely on keywords or metadata, fall short in capturing the nuanced relationships between visual and textual information required for such applications.

Composed Image Retrieval (CoIR) is a prominent example of this approach, where a user provides a reference image and a textual modification to search for a target image. For instance, given a picture of a blue car and the query "change the color to red," the system retrieves images of red cars. This paradigm has proven successful in static image contexts but does not address the additional complexity introduced by video data, such as motion, temporal dependencies, and multi-frame consistency.

Composed Video Retrieval (CoVR) builds on CoIR by extending its capabilities to videos. In CoVR, the query consists of a reference video and a textual modification. For example, given a video of a person walking and the query "make them run," the system retrieves videos depicting the same scene but with the person running instead of walking.

This requires the system to process multimodal inputs insofar as CoVR uses pre-trained vision-language models, such as BLIP [3] or BLIP-2 [4], to encode both video and textual inputs into a shared latent space. Moreover, videos are treated as sequences of frames. To simplify processing, a representative embedding for each video is computed, typically by encoding key frames or averaging embeddings across frames. The system learns a joint embedding space where the modified query (video + text) is projected close to the target video embedding. This alignment is achieved through contrastive learning, a technique that minimizes the distance between positive pairs (query and target) while maximizing the distance from negatives. Unlike traditional approaches that rely on costly manual annotations, CoVR-2 employs an automatic pipeline to generate triplets of videos and captions.

2. Problem Statement

The task of CoVR can be formalized as follows: Given a reference video q , a textual modification t , and a video database $B = \{v_1, v_2, \dots, v_n\}$, the goal is to retrieve the target video v_k that aligns most closely with the modified query $f(q, t)$. Here, f represents a learnable function mapping the query-text pair to a shared latent space. The target video embedding $h(v_k)$ is learned to be close to $f(q, t)$ in this space, using contrastive learning to minimize the distance between positive pairs while maximizing it for negatives.

The pipeline begins by identifying pairs of videos from a large corpus, each paired with a caption describing its content. The MTG-LLM (Modification Text Generator LLM) analyzes these video-caption pairs to infer semantic differences and generates a textual description of the modifications required to transform one video into the other. For instance, given a video-caption pair "a person walking in the park" and another "the same person running in the park," the MTG generates the modification text "change the action to running." Using this approach, the triplets consist of the reference video, the modification text describing the semantic change, and the target video reflecting the described change. The resulting datasets, CC-CoIR and WebVid-CoVR, serve as benchmarks for training for respectively CoIR and CoVR models and WebVid-CoVR-test for evaluating CoVR models, enabling effective learning without manual effort.

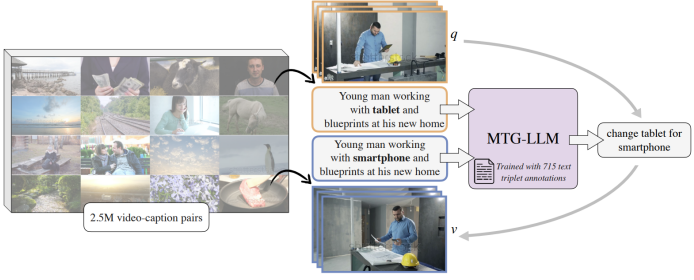


Figure 2: Triplet generation method overview

By leveraging web video-caption pairs, the authors generate diverse triplets that form the foundation of the dataset. Each triplet consists of a reference video, which serves as the starting point, a textual modification describing the specific changes to be applied, and a target video that reflects the described modifications. This structure enables the model to learn the relationship between the reference, the textual description, and the resulting target video, ensuring a comprehensive representation of multimodal queries.

The CoVR-2 framework leverages the BLIP-2 model which integrates two primary components to process and align diverse data modalities. The first component, the Q-Former, is a query-centric transformer that encodes textual and visual inputs, enabling complex interactions between the two modalities. The second component is the vision-and-language encoder, which combines image and text features into a shared latent space, ensuring effective alignment for downstream retrieval tasks. Together, these components enable BLIP-2 to handle the intricate relationships between visual and textual data with high precision.

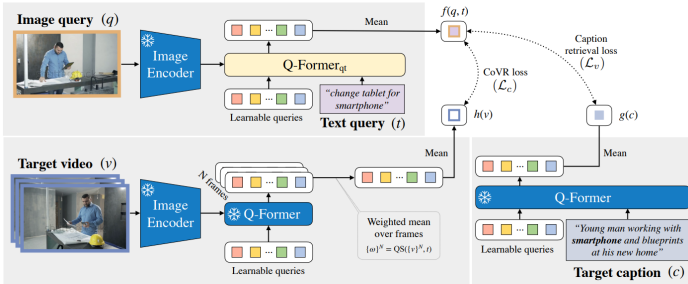


Figure 3: CoVR-BLIP2 architecture overview

BLIP-2 has demonstrated strong performance on the Composed Image Retrieval (CoIR) task, showing state-of-the-art results on benchmarks such as CIRR.

3. Our Works

For evaluating the CoVR-2, we conducted our experiments using an **NVIDIA V100 GPU with 32 GB of memory**, supported by a **76 GB of RAM CPU** and 400 GB of disk storage for datasets and intermediate results. The training was performed with a batch size of 32 over two epochs for initial evaluations, using the AdamW optimizer and a learning

rate of 1×10^{-4} for the BLIP-2 backbone. For evaluation, we used standard metrics such as Recall@K (R@K) to measure retrieval performance. Additionally, we reproduced the baseline results from the [Github](#) for BLIP-2 on the CIRR dataset, adjusting batch size and epochs to accommodate resource limitations, providing a consistent foundation for comparing our improvements.

3.1. Reproducing paper results

We began by reproducing the baseline results using the CoVR-BLIP-2 model, utilizing a checkpoint trained on the configuration file `blip2-l-cirr-coco-coir`. This checkpoint provided a pretrained model specifically tailored for the CIRR dataset. Additionally, we conducted training from scratch using the same dataset and configuration, allowing us to compare the performance of pretrained and freshly trained models. While the checkpoint model delivered strong baseline results, our from-scratch training, conducted with limited computational resources, provided comparable performance, serving as a robust foundation for subsequent experiments.

Method	Ckpt	Recall@K			
		K=1	K=5	K=10	K=50
BLIP2 [1]	-	50.87	80.80	88.84	98.00
BLIP2 [1]	✓	50.63	81.04	89.35	98.15
BLIP2 (our)	-	30.24	59.74	71.81	90.89
BLIP2 ref. (our)	✓	34.34	63.93	74.55	92.46

Table 1: Results reproduction on the CIRR test set

3.2. Balancing Embedding Contributions

By applying weighting to the text, image, and multi-modal embeddings, the goal was to fine-tune the contributions of each component, ensuring a balanced and context-aware representation. This strategy addresses the varying importance of each modality in different scenarios—where the image, text, or their combination might carry more significance.

To study the optimal combination of image, text, and multi-modal embeddings, we conducted two distinct experiments. In the first experiment, we used a fixed-weight approach where each modality contributed equally to the final representation. The final embedding was computed as a simple average:

$$\frac{1}{3} \cdot \text{img_embd} + \frac{1}{3} \cdot \text{txt_embd} + \frac{1}{3} \cdot \text{multi-modal_embd}$$

In the second experiment, we implemented a multi-layer perceptron (MLP) to dynamically adjust the contributions of the embeddings. The MLP was trained to learn optimal weights, w_1, w_2, w_3 , for each modality, producing the final embedding as:

$$w_1 \cdot \text{img_embd} + w_2 \cdot \text{txt_embd} + w_3 \cdot \text{multi-modal_embd}$$

The MLP consists of fully connected layers interleaved with non-linear activation functions, enabling it to capture

complex interactions between the embeddings. However, our results indicate that neither the fixed-weight approach nor the MLP-based approach outperformed the baseline.

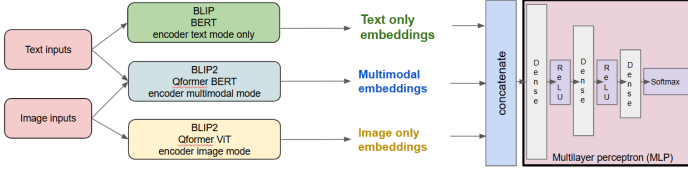


Figure 4: MLP Balancing Embedding Contributions

The fixed-weight and MLP-based weighting approaches likely underperformed. The fixed-weight method assumes equal contributions from all modalities, which may not align with the varying importance of image, text, and multi-modal embeddings for specific queries

Meanwhile, the MLP-based approach likely underperformed because it overfit to the multi-modal embedding, disproportionately emphasizing it over the individual image and text embeddings. This overfitting occurred because the multi-modal embedding, being a combined representation, already contains significant information from both modalities. As a result, the MLP may have learned to rely heavily on this embedding while neglecting the unique contributions of the separate image and text embeddings. This imbalance reduced the model’s ability to generalize effectively, as it became overly reliant on a single source of information rather than leveraging the full diversity of the embeddings.

Method	Recall@K			
	K=1	K=5	K=10	K=50
BLIP2 ref. (our)	34.34	63.93	74.55	92.46
Equal weights (our)	23.98	51.52	64.17	87.64
MLP (our)	33.57	63.52	74.87	92.72

Table 2: Scores with different balancing embedding contributions on the CIRR test set

3.3. Consistency Loss Integration

To address the overfitting issue of the MLP, we introduced a consistency loss between the image, text, and multi-modal embeddings. This loss encourages the embeddings to align more closely in the shared latent space, ensuring that the multi-modal representation complements the individual embeddings rather than dominating them.

We will use the consistency loss function defined in the paper [1]:

$$L_v(B) = - \sum_{i \in B} \log \left(\frac{e^{S_{i,i}/\tau}}{\alpha \cdot e^{S_{i,i}/\tau} + \sum_{j \neq i} e^{S_{i,j}/\tau} w_{i,j}} \right) - \sum_{i \in B} \log \left(\frac{e^{S_{i,i}/\tau}}{\alpha \cdot e^{S_{i,i}/\tau} + \sum_{j \neq i} e^{S_{j,i}/\tau} w_{j,i}} \right)$$

The first component, si-img_loss, enforces alignment between the query embedding and the target video embedding. The second component, si-text_loss, aligns the query embedding with the text embedding, enabling the model to incorporate semantic information from textual modifications into the representation. Finally, si-ti_loss, ensures that the query embedding aligns closely with the target embedding while maintaining sufficient separation from negative samples.

These components were integrated into the initial loss framework proposed by the authors and further augmented with additional loss calculations to enhance retrieval robustness. While the original loss served as the foundation, we extended it to better capture the nuances of multimodal alignment. Specifically, our augmentation introduced additional regularization and weighting strategies to address the varying importance of modalities, ensuring that the final representation leveraged information from all embeddings effectively.

We conducted two experiments to evaluate the impact of different weighting strategies in the consistency loss. First, we used an equi-balancing consistency loss where each component was weighted equally. In the second, we applied a weighted loss scheme where the si-ti_loss was given twice the weight of the si-img_loss and si-text_loss. Our findings indicate that the weighted loss strategy, where si-ti_loss had a higher importance, outperformed the equi-balancing approach for low value of K, although it still fell short of reaching the baseline performance.

Future work could address this issue by introducing additional regularization, imposing constraints on the learned weights, or incorporating attention mechanisms to enhance the robustness of dynamic weighting strategies

Method	Recall@K			
	K=1	K=5	K=10	K=50
BLIP2 ref. (our)	34.34	63.93	74.55	92.46
Equi loss balancing (our)	26.34	54.31	66.39	89.47
Fixed loss balancing (our)	27.04	55.1	66.87	89.18

Table 3: Performance comparison of different consistency loss weighting strategies on CIRR test set

4. Conclusion

The CoVR-2 framework provides a versatile and robust foundation for advancing Composed Video Retrieval tasks through automatic dataset construction and innovative multimodal integration strategies. The methods explored in this study are highly generalizable and open several avenues for future extensions. Further refinements, such as incorporating adaptive weighting strategies or attention mechanisms, could enhance alignment even further. Our experiments revealed that the choice of loss weighting schemes and embedding strategies influences retrieval accuracy, suggesting the potential for active learning methods where loss weights or other hyperparameters dynamically evolve during training.

5. Acknowledgements

We would like to express our sincere gratitude to Lucas Ventura for his mentoring and his invaluable support during this project. Lastly, we extend our thanks to the teachers of the class RecVis24, which made possible the execution of our experiments.

References

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